

cracking when the board is subjected to high temperature.

Copper foils come in different thicknesses. The most common are given in the table below.

Use of weight indications reflects the weight in ounces (oz) of 929.03sqcm (1-square-foot) of copper foil. (For practical reasons, it is stated as ounce.) Aluminium-based substrates are available with 17.5µm, 35µm, 70µm and 105µm copper foils.

Dielectric layer. This substance electrically isolates the aluminium base layer from the copper foil and allows for rapid heat transfer between the two. It ensures that heat generated in the components is dispersed to the heat-sink as quickly as possible, and determines the materials' thermal properties.

Dielectric layers are made of special polymer filled in ceramics. These are thermally conductive, having minimum thermal resistance. Each dielectric layer has its own Underwriter Laboratory (UL) recognition.

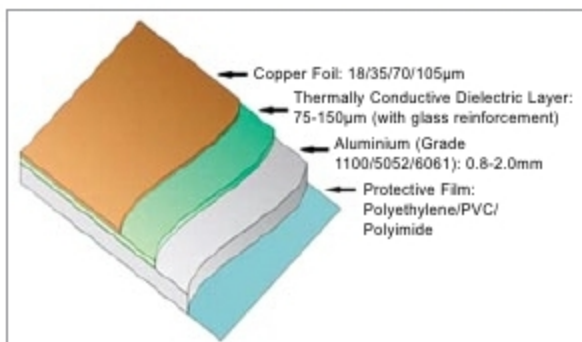


Fig. 2: Layers of insulated metal substrate (Credit: CCI, EUROLAM)

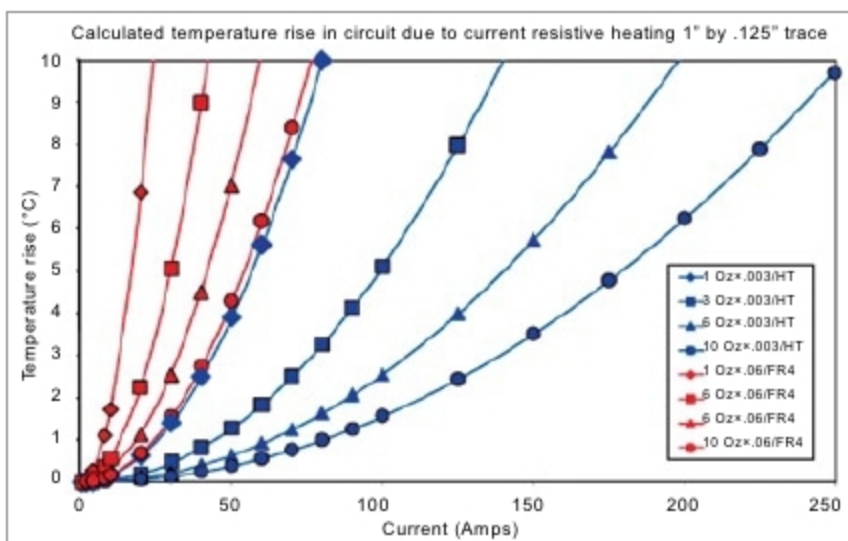


Fig. 3: Thermal conductivity of different IMSes

DIFFERENT TYPES OF COPPER FOILS

	Type I	Type II	Type III	Type IV	Type V	Type VI	Type VII
Thickness (µm)	5	9	12	17.5	35	70	105
Weight (oz)	1/8	1/4	1/3	1/2	1	2	3

Dielectric insulating material in metal-clad materials is designed to have a much superior thermal conductivity than that of FR4—generally ranging from 1W/m-K to 9W/m-K—allowing for efficient heat removal.

Thickness of dielectric layers also varies from 75µm to 100µm and 150µm. Other properties that may help material selection are dielectric thickness, maximum operating temperature (MOT), breakdown voltage and peel strength.

Based on the dielectric layer, metal-clad printed circuit board (MCPCB) base materials are differentiated.

Aluminium base. Aluminium is widely used as a heat-sink because of its low cost and good thermal conductivity. Thermal conductivity of aluminium is 2.3W/cm-K. Although thermal conductivity of aluminium is not as good as that of silver, gold or copper, due to cost factor, aluminium is widely used as a heat-dissipating component.

Even though thermal conductivity of copper is better than that of aluminium, the latter can radiate heat into the air better than the former, because of its low density. So, in most cases, base layer is aluminium—generally 0.8mm - 3.2mm thick (most common is 1.6mm). Aluminium is an efficient conductor of heat and, as such, perfect for applications involving rapid heat transfer.

Protective film. This is a self-sticking thin film placed at the bottom of the aluminium PCB base material.

It protects the material from handling damage as well as protects the aluminium base at the time of etching during PCB production.

Thermal conductivity. This is a measurement of the ability of a substance to conduct heat, measured in W/mK (watts per metre Kelvin).

Benefits of using MCPCBs are given below.

1. Improved heat dissipation without using external heat-sinks
2. Mechanically rigid base laminates

3. Dramatically reduced thermal stress on electronic components, improving reliability and mean time between failure (MTBF)

4. Lower operating temperatures, longer component life and increased durability

5. Better thermal placement that permits closer components mounting and smaller PCBs

6. Fewer heat-sinks and other mounting hardware, including thermal interface material

7. Possibility of using surface-mount power and passive devices

8. Relatively low cost

9. Better thermal management that allows more forward current to be applied to LED, which means more light and possibly a lower number of LEDs for desired light output; maintaining a cooler assembly at equivalent power equates to more light per die

Types of IMS

IMS comes in a variety of types, including HPL-03015, HT-04503, HT-07006 and MP-06503.

HPL-03015. High power lighting (HPL) is a dielectric specifically formulated for high power lighting LED applications with demanding thermal performance requirements. This thin dielectric at 38µm (0.0015-inch) has the ability to withstand high temperatures with a glass transition of 185°C and phenomenal thermal performance of 0.30°C/W.

HT-04503. High temperature (HT) 3-mil is a dielectric resistant to degradation from high temperature exposure. It features high dielectric breakdown characteristics. This dielectric is proven in applications such as LEDs, power conversion, heat-rails, solid-state relays and motor drives.

HT-07006. HT 6-mil is a dielectric resistant to degradation from high temperature exposure. It features higher dielectric breakdown characteristics than its 3-mil counterpart. This dielectric is proven in applications such as LEDs, power conversion, heat-rails, solid-state relays and motor drives.

MP-06503. Multi-purpose (MP) is an industry-proven dielectric for a multitude of applications including LEDs, power conversion, heat-rails, solid-state relays and motor drives. **EFY**